

VM types: VMAS (Availability Set) and VMSS (Scale Set)

- VMAS



- Collection of discrete virtual machines that assist in preventing the impact of a single point of failure or a maintenance event by dispersing VMs across various fault domains and update domains
- Update domains refer to the number of VMs that can be updated concurrently during planned maintenance performed by Microsoft. A maximum of twenty (20) update domains exist
- A fault domain contains VMs sharing the same power source and network switch, limiting the impact of potential network or power outages and physical hardware failures. A maximum of three (3) fault domains exist

- VMSS



- Collection of identical virtual machines that are deployed and maintained together
- Provide automatic VM scaling and management, allowing for the quick addition or removal of virtual machines in response to changing demand
- Default set type in AKS

Main VMAS limitations in AKS:

- You cannot have multiple node pools with VMAS
- You cannot enable Cluster Autoscaler

Main VMSS limitations in AKS:

- They are deployed in a single fault domain